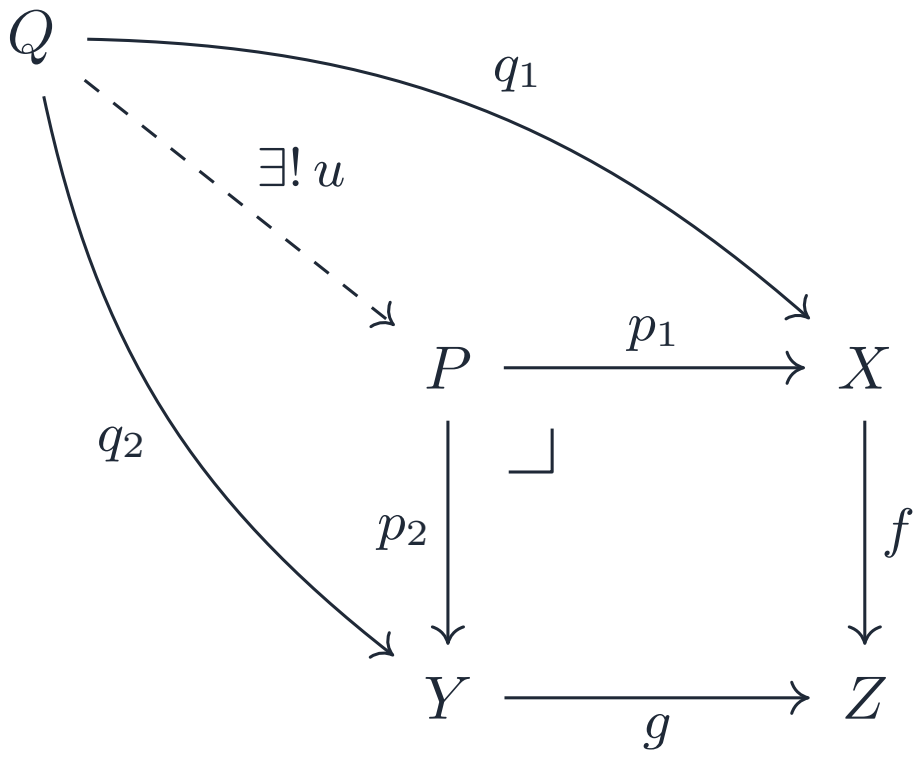
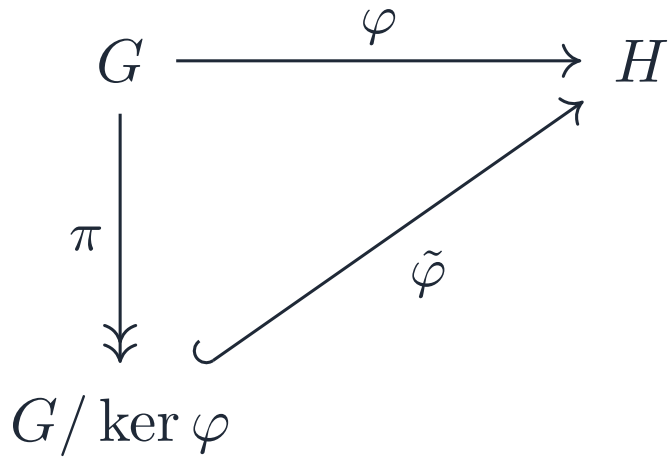




$$G/\ker\varphi \;\cong\; \operatorname{im}\varphi$$